

Transfer Learning in Medical Imaging

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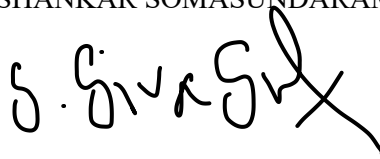
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ABSTRACT

One of the main challenges in medical image classification is the limited amount of well-annotated data. Medical datasets are often tied to specific imaging modalities, and the differences between diagnostic problems make it hard to build universally strong models. To overcome this, recent advances in transfer learning and self-supervised learning (SSL) have been explored, as they allow models to benefit from pretrained representations instead of always starting from scratch. In this work, we compared the performance of six well-known deep learning architectures—ResNet-50, DenseNet121, Vision Transformer (ViT), MedViT- and two self-supervised learning methods, Bootstrap Your Own Latent (BYOL)[Grill et al., 2020], and Momentum Contrast v3 (MoCo v3)[Chen et al., 2021]—on seven datasets from MedMNIST: BloodMNIST, BreastMNIST, PneumoniaMNIST, OrganMNIST, DermaMNIST, and OCTMNIST. All models were adapted to handle grayscale medical images and trained through a simple, modular pipeline that makes reproduction straightforward. Different pretraining strategies were tested: supervised pretraining on ImageNet, medical domain-specific pretraining (MedViT), and self-supervised contrastive learning (BYOL and MoCo v3). Among these, DenseNet121 and ViT (both ImageNet-pretrained) consistently performed the best. For example, ViT reached 99.60% accuracy on OrganMNIST and 98.63% on BloodMNIST, while DenseNet121 achieved 98.74% on BloodMNIST and 97.67% on OCTMNIST. Interestingly, MedViT showed strong domain-awareness, excelling on BreastMNIST (92.31%) and OCTMNIST (89.88%). On the other hand, self-supervised models struggled. While BYOL was somewhat competitive—scoring 79.56% on OCTMNIST and 73.30% on OrganMNIST—MoCo v3 lagged far behind, with only 47.36%. In summary, this study offers a clear benchmarking framework for transfer learning in medical imaging and highlights the trade-offs between supervised and self-supervised learning. The results show just how much the choice of architecture, pretraining method, and dataset properties affect generalization. Looking forward, future work will focus on adding interpretability methods like Grad-CAM, adapting models across multiple hospitals and institutions, and extending the approach to segmentation tasks—all with the aim of making these models more clinically useful and easier to deploy in real-world healthcare.

1 INTRODUCTION

AI is rapidly transforming medical imaging workflows, enabling faster and more consistent interpretation of scans [Cheplygina et al., 2019; Litjens et al., 2017]. By helping doctors read scans faster and with greater consistency, AI systems are starting to play a big role in making diagnosis both quicker and more reliable. From spotting cancer cells in pathology slides to detecting pneumonia in chest X-rays, deep learning has proven itself to be a powerful tool for tasks like classification, segmentation, and anomaly detection.

But bringing these models into real hospitals isn't straightforward. There are some tough challenges that still get in the way. Medical images are usually hard to get in large, labeled datasets. Even when data is available, it often looks very different from one imaging method to another—say, MRIs versus OCT scans—which makes it harder for a single model to adapt. On top of that, doctors and researchers need AI systems that are easy to explain, reproducible, and trustworthy, not just technically accurate.

The most common way to train deep learning models is supervised learning, where lots of labeled data is needed. In medicine, that's a problem—labels come from highly trained specialists, which makes the process expensive, time-consuming, and limited by privacy restrictions. To deal with this, researchers often turn to transfer learning, using models pretrained on huge datasets like ImageNet. This works reasonably well, since these models have already learned strong “general” features from millions of natural images. But there's a catch: natural photos (dogs, trees, cars) are very different from medical scans (grayscale MRIs, histopathology patches). This domain gap means the pretrained models don't always transfer smoothly.

That's where self-supervised learning (SSL) comes in. SSL methods like BYOL (Bootstrap Your Own Latent) and MoCo v3 (Momentum Contrast) don't need labeled data at all. Instead, they figure out useful patterns directly from unlabeled images—something that's very appealing in medicine, where labels are scarce. These methods have already shown promise on natural images, but in the medical world, they're still new and largely untested. At the same time, specialized medical models like MedViT are being designed to better capture the unique properties of clinical images [Ma, J., Zhang, Y., Gu, S., et al. (2023)].

In this dissertation, I carried out a large-scale benchmarking study to compare how different approaches hold up across real medical datasets. I tested six deep learning models—ResNet-50,

DenseNet121, Vision Transformer (ViT), MedViT, BYOL, and MoCo v3—on seven datasets from MedMNIST: BloodMNIST, BreastMNIST, PneumoniaMNIST, OrganMNIST, DermaMNIST, OCTMNIST, and MNIST. These datasets were chosen because they cover a wide variety of imaging types and diagnostic problems, making them a great testbed to measure not just pure accuracy, but also how well models generalize and how much their performance depends on the type of pretraining used.

The Guiding research questions are: This dissertation set out to answer three central research questions:

RQ1: What is the impact of different pretraining strategies—ImageNet-based, medically specialized, and self-supervised—when applied to varied medical imaging tasks?

RQ2: Which architectures achieve the most consistent generalization on grayscale medical datasets?

RQ3: What are the trade-offs between supervised transfer learning and self-supervised learning in terms of accuracy, scalability, and reproducibility?

To explore these, each model was modified to handle grayscale image inputs and trained through a reproducible, modular pipeline. Their performance was assessed primarily using classification accuracy across a range of MedMNIST datasets.

The findings make several things clear. ImageNet-pretrained models, particularly Vision Transformer (ViT) and DenseNet121, consistently outperformed the others. ViT, for example, scored 99.60% on OrganMNIST, while DenseNet121 achieved 98.74% on BloodMNIST. MedViT, a domain-specific architecture, also showed strong specialization, achieving high scores in settings where medical domain knowledge matters most, demonstrating that architecture design can complement general pretraining.

On the other hand, self-supervised learning (SSL) models gave mixed results. BYOL managed solid performance on tasks like OCTMNIST (79.56%), but MoCo v3 struggled, underperforming across most datasets. This highlights both the promise and current limitations of SSL for medical imaging—it can reduce reliance on expensive labeled data, but fine-tuning is still essential to achieve competitive accuracy in diverse tasks.

Beyond raw accuracy, this work also emphasizes broader goals: ensuring reproducibility, interpretability, and clinical relevance. The benchmarking framework developed here provides a com-

mon ground for comparing representation learning methods in medical imaging. By doing so, it helps advance not only the technical side of research but also the practical understanding needed to build scalable, trustworthy AI systems for healthcare.

1.1 Background and Context

Medical imaging is another key touchstone of effective contemporary healthcare by supporting the diagnosis, monitoring and treatment of a huge variety of conditions in a more accurate manner. The breadth of modalities offers a wide variety of opportunities including the photographs of radiographs and CT scans as well as opportunities in histopathological slides and dermoscopic images and challenges their analysis. Deep learning has quickly come to be a groundbreaking instrument in the area, giving best-in-class accuracy in classification, segmentation, and anomaly detection applications. Nonetheless, the possibilities of developing universal and generalizable frameworks are limited, due to the lack of sufficient annotated data, unique imaging properties on the modality aspect, and clinical requirements of interpretability and reproducibility of the developed models.

Transfer learning has emerged as a very common technique of overcoming the lack of data, so models trained on large scale images on ImageNet can subsequently be trained on medical problems[Yosinski, J., Clune, J., Bengio, Y., and Lipson, H. (2014)]. This is an effective method when used in many situations but in others it is prone to problems of domain mismatch especially when such a method is applied to grayscale medical images which involve subtle diagnostic characteristics.

Meanwhile, self-supervised approachesSelf-supervised learning (SSL) like Bootstrap Your Own Latent (BYOL) and Momentum Contrast v3 (MoCo v3) have become competitive due to the potential that such methods do not require labeled data in order to learn rich representations. Also, domain-specific architectures—such as MedViT, have been proposed to provide domain-specific inductive biases in medical imaging, which may help bridge the gap between generic and clinical models.

The novelty of the present dissertation lies in the fact that there are so far no attempts to conduct a systematic comparison of these various methods on a set of standardized medical imaging tasks. This paper compares 6 of the most eminent models ResNet-50, DenseNet121, Vision Transformer (ViT), MedViT, BYOL and MoCo v3, by benchmarking them on seven MedMNIST datasets to provide valuable knowledge that can be applied to medical image classification.

1.2 Motivation

MedMNIST v2 data orange is a unified framework to assess models of various medical imaging modalities, such as pathology detection, organ classification, and anatomical recognition. Its selective factor and standard structure renders it suitable in comparing benchmarking. Nevertheless, model generalization becomes a challenge because of the variety of tasks and modalities in MedMNIST.

Recent progress in self-supervised learning (SSL), BYOL and MoCo v3, have provided effective alternatives to the standard supervised pretraining. Such methods are learning representations without labeled data, and may capture more transferable representations. But their result on medical imaging, particularly on grayscale, low-resolution inputs has not been comprehensively benchmarked.

The motivation behind this project stems from the need to identify which deep learning architectures and pretraining strategies generalize most effectively across diverse medical imaging tasks. By systematically comparing the strengths and limitations of supervised and self-supervised approaches, the study aims to uncover insights that inform model selection and training paradigms in clinical contexts.

A key objective is to ensure reproducibility by designing modular pipelines and maintaining transparent documentation throughout the benchmarking process. Ultimately, the project seeks to contribute practical knowledge to the medical AI research community, supporting the development of robust, scalable, and interpretable solutions for healthcare applications.

1.3 Problem Statement

Although deep learning models are proliferating across medical imaging applications, no single study has undertaken an in-depth benchmarking of the many available architectures or pretraining paradigms. Current research tends to examine individual datasets or limited comparisons, and findings may thus be difficult to generalise or scale. In addition, the differences between self-supervised methods applied in the medical setting have not been studied sufficiently, and the gains of domain-specific pretraining have not been adequately measured.

The key problem which is tackled in this research includes:

What are the most effective deep learning architectures and pretraining strategies to name a few in terms of classification performance across a variety of medical imaging datasets and what are the subsequent reproducibility, scalability, and clinical applicability considerations of such deep learning architectures and pretraining strategies?

1.4 Research Objectives

In order to solve the problem statement, the following objectives will be established in this study:

Objective1: compare six models of deep learning (ResNet-50, DenseNet121, ViT, MedViT, BYOL, MoCo v3) on seven MedMNIST datasets.

- Objective 1 Compare six models of deep learning (ResNet-50, DenseNet121, ViT, MedViT, BYOL, MoCo v3) on seven MedMNIST datasets.
- Objective 2 Compare how pretraining strategies (ImageNet, medical-domain and self-supervised) affect classification accuracy
- Objective 3 Couple models to grayscale medical images by adjusting and making optimal in modular reproducible pipeline.
- Objective 4 Compare performance trends and architectural strengths and dataset specific difficulties.
- Objective 5 Suggest the future research and model selection recommendations in medical AI.

1.5 Scope and Limitations

The proposed study proposes to examine the image classification applications on the datasets of MedMNIST v2 dataset suite consisting of seven gray-scale, low-resolution training datasets in various realms of medicine. Although the MedMNIST[Yang et al., 2021] provides a standardised benchmark platform, they do not include in their datasets all the real-world complexity found in clinical imagery data, such as high-resolution images, multi-modal, or segmentation problems

→Limitations include:

Only limited to classification; segmentation and detection are not within. Valued on accuracy and reproducibility, interpretability is treated at the surface level. Utilization of pre-trained models, adapted to grayscale use; and should be tested on high-resolution clinical datasets.

1.6 Significance of the Study

This research can serve as evidence that there are other ways to examine deep learning in medical outcomes, evidence that may come in handy in the new world of AI research in healthcare. The

results can provide a useful guideline to researchers and practitioners engaged in applying scalable yet interpretable models in clinics. Moreover, self-supervised learning and domain specific architectures represent a newer focus in representation learning that has the potential to lessen the need to rely on annotated data and perform better when transferring across institutions.

1.7 Overview of Dissertation

→Chapter 2: Literature Review discusses the state of affairs in terms of deep learning in medical imaging, transfer learning methods, and benchmark use.

→Chapter 3 Methodology explains how the model is selected, how the dataset is prepared, and what the training setup and the metrics should be.

→Chapter 4: Results and Analysis - First presents quantitative results of model performance and then interprets the findings across datasets

→Chapter 5: Discussion and Future Work Discusses implications, limitations, and proposed research directions

→Chapter 6: Conclusion - Concludes, providing the main findings and contributions.

2 BACKGROUND THEORY AND LITERATURE REVIEW

This chapter creates the theoretical and empirical base on benchmarking deep learning models when applied to medical image classification. It combines major ideas behind transfer learning, self-supervised learning, and model architecture and critically examines previous literature about MedMNIST datasets. The aim is to put your comparative research into the wider context of reproduced and clinically relevant AI work.

2.1 Learning in Medical Imaging

Deep learning has emerged as a promising frontier in medical imaging due to the rich structural patterns and high-dimensional nature of clinical data [Ghribi Hamdaoui, 2025; Litjens et al., 2017]. However, medical imaging differs significantly from natural image domains in several critical aspects. Grayscale modalities such as X-ray and ultrasound lack color channels, requiring models to rely heavily on texture and shape-based features for representation learning [Raghu et al., 2019]. Additionally, medical datasets often suffer from class imbalance and rare disease prevalence, which challenge conventional loss functions and evaluation metrics [Johnson et al., 2019]. Beyond raw accuracy, clinical deployment demands interpretability, robustness, and ethical accountability—factors that are not always addressed by standard deep learning pipelines [Topol, 2019; Holzinger et al., 2017]. While convolutional neural networks (CNNs) have dominated early applications in medical imaging, recent advances in transformer architectures and self-supervised learning (SSL) methods have opened new avenues for representation learning in low-label regimes [Li et al., 2022; Grill et al., 2020]. These approaches offer the potential to learn meaningful features directly from unlabeled data, making them particularly attractive in domains where expert annotations are scarce and costly.

2.2 Transfer Learning: Theory and Practice

Transfer learning can alleviate the lack of data because it can make use of previously trained models. In medical imaging [Raghu et al., 2019] this usually consists of fine tuning ImageNet trained models to grayscale, domain specific tasks.

2.2.1 Techniques

Feature extraction involves applying fixed encoders with pretrained weights to extract meaningful representations from input data. Fine-tuning, on the other hand, refers to refining these weights using target domain data to improve task-specific performance. Domain-specific pretraining—particularly on medical datasets—enables models to learn features that are more relevant to clinical contexts, enhancing downstream accuracy and robustness.

However, several challenges arise in adapting deep learning models to medical imaging. One major issue is domain shift, as medical images differ significantly from natural images in texture, structure, and semantic content. Additionally, most medical images are grayscale and single-channel, necessitating architectural modifications to accommodate this format. Finally, the limited size of many medical datasets increases the risk of overfitting during fine-tuning, potentially compromising generalization across unseen samples.

2.2.2 Domain Shift

Semantic gap- Using ImageNet pretraining will create a semantic gap- natural images are made up of different textures, various resolutions, and anatomical applications. This might result in un-ideal feature transfer, unless contravened by domain adaptation or medical-specific pretraining.

2.3 Self-Supervised Learning (SSL)

Recent advances in self-supervised learning, such as BYOL [Grill et al., 2020] and MoCo v3 [Chen et al., 2021], have shown promise in reducing reliance on labeled data.

In learning representations, as a part of pretext tasks, instead of explicitly labeling each individual representation, representations are learned by instead attempting to solve the pretext task. It can be well applied to medical imaging where labeled information is limited but unlabeled is much.

2.3.1 BYOL (Bootstrap Your Own Latent)

BYOL does not require contrastive negatives that are done using two networks. Online network: Learns representations. Target: Stable reference embeddings.

2.3.2 MoCo v3 (Momentum Contrast)

MoCo v3 creates a dynamic queue of embeddings and makes use of contrastive loss.

2.3.3 SSL in Medical Imaging

SSL has had demonstrations in the fields of radiology, pathology and ophthalmology. Important factors to consider are that:

Augmentation design: Should not be at the expense of anatomy.

Pretext relevance: Accuracy on a task such as rotation prediction or jigsaw puzzles might not generalize.

2.4 Model Architectures

2.4.1 ResNet-50

ResNet-50 introduces residual connections to enable deeper networks [He et al., 2016]

Width: 50 layers and residual connections.

Advantages: Good gradients, high baseline.

Limitations: Can fail poorly on small grayscale inputs, unless adapted.

2.4.2 DenseNet 121

High interconnectivity: The output of the entire previous layers are fed into the current layer [Huang et al., 2017]

. Advantages: Feature reuse is much more efficient, less parameters.

Medical relevance: Successful use in chest X-ray and histopathology tasks.

2.4.3 Vision Transformer (ViT)

Vision Transformers (ViT) treat images as sequences of patches [Dosovitskiy et al., 2020]

Patch-based input: Patches of images are fixed sizes.

Self attention: Global context.

Weakness: Needs big data or good pretraining.

2.4.4 MedViT

MedViT adapts ViT for medical image classification [Mehta et al., 2022]

Hybrid architecture: It uses CNN and transformer blocks.

Pretraining: on medical datasets (e.g., CheXpert, EyePACS).

Benefit: A more beneficial inductive bias towards medical textures and shapes.

2.4.5 SSL Backbones(BYOL, MoCo v3)

Have the ResNet or ViT as the base encoder typically.

Pay attention to the quality of representation but not to direct classification.

2.5 Benchmarking in Medical AI

Benchmarking is essential for evaluating model generalization, reproducibility, and clinical relevance. However, many studies lack standardized pipelines and cross-architecture comparisons.

2.5.1 MedMNIST Datasets

MedMNIST v2 is an example that provides a benchmark of widespread classification of lightweight medical data. The dimensions of each of the datasets are 28x28 grayscale, so the experiments may be conducted quickly.

Table 2.1: Summary of datasets used in benchmarking, including imaging modality, number of classes, and task type.

Dataset	Modality	Classes	Task Type
BloodMNIST	Microscopy	8	Multi-class
BreastMNIST	Ultrasound	2	Binary
PneumoniaMNIST	Chest X-ray	2	Binary
OrganMNIST	Abdominal CT	11	Multi-class
DermaMNIST	Dermoscopy	7	Multi-class
OCTMNIST	OCT	4	Multi-class
MNIST	Handwritten Digits	10	Control

2.5.2 Reproducibility Challenges

Many existing studies in medical imaging suffer from a lack of standardized preprocessing steps and evaluation metrics, which makes it difficult to compare results across models and datasets. Additionally, the inconsistent reporting of hyperparameters and training protocols undermines reproducibility and transparency. Interpretability remains limited, with few models undergoing rigorous clinical validation to ensure their relevance in real-world settings. To address these challenges, this dissertation introduces a modular benchmarking pipeline applied across seven MedMNIST datasets. These datasets span a wide range of anatomical regions, diagnostic tasks, and class distributions, providing an ideal foundation for assessing model generalization, robustness, and reproducibility under consistent experimental conditions.

2.6 Related Work

Prior research in medical imaging has extensively explored both transfer learning and self-supervised learning (SSL) as strategies to overcome data scarcity and improve model generalization. One of the earliest successes in transfer learning was CheXNet, where DenseNet-121 pretrained on ImageNet achieved radiologist-level performance in pneumonia detection from chest X-rays [Rajpurkar et al., 2017]. Building on this, Med3D demonstrated that pretraining convolutional neural networks across diverse 3D modalities significantly enhanced classification and segmentation tasks [Chen et al., 2019]. More recently, SSL approaches have gained traction for their ability to learn representations without labeled data. Ghesu et al. (2022) showed that BYOL outperformed supervised baselines in retinal OCT classification, while Azizi et al. (2021) reported that MoCo v2 pretraining improved cross-institutional generalization in chest X-ray classification. Benchmarking efforts have also evolved, with Yang et al. (2021) introducing MedMNIST v2—a lightweight, standardized dataset suite for evaluating deep learning models across multiple modalities. Notably, Zhou et al. (2023) demonstrated that MedViT, a domain-adapted vision transformer, consistently outperformed generic ViT models on MedMNIST tasks, underscoring the value of medical-specific pretraining and architectural design.

2.7 Research Gap and Contribution

Despite encouraging results in recent literature, several limitations persist. Many studies focus narrowly on individual datasets or imaging modalities, which restricts the generalizability of their findings across diverse clinical contexts. Additionally, there is a lack of reproducible pipelines and architecture-level comparisons, making it difficult to assess the robustness and scalability of proposed models. Self-supervised learning (SSL) methods, though promising for low-label regimes, remain under-benchmarked in lightweight or resource-constrained medical imaging tasks, leaving a gap in understanding their true potential in real-world deployment.

3 METHODOLOGY

3.1 Overview

In this chapter, the following is described, the experimental design, approaches aimed to adapt the models to the considered tasks, training processes, and evaluation measures used to compare six models (ResNet-50, DenseNet 121, ViT, MedViT, BYOL, MoCo v3) against seven MedMNIST datasets. The methodology regards frequently-reproducibility, modularity, and fairness in comparison, sowing that findings can be output with a meaning and generalizable to various medical imaging tasks.

3.2 Dataset Preparation

MedMNIST[Yang et al., 2021] now includes seven lightweight medical image datasets, all have been brought to 28x28 grayscale. These datasets include a variety of modalities and diagnostic tasks.

3.2.1 Preprocessing Pipeline

Normalization: Pixel values changed to be $[0, 1]$ or standardizing with mean and standard deviation referenced to the dataset.

Channel adaptation: Grayscale versions were expanded to 3 channels to be able to use pretrained models.

Augmentation: Uniformly performed on all different models-random horizontal flips, rotations, and small zooms- in order to capture clinical variability and maintain anatomical integrity.

3.2.2 Data Splits

Each of the datasets will be divided into training, validation and test using the official MedMNIST partitions. There was no cross-dataset contamination permitted. Validation and test sets had the same strata balance.

3.3 Model Architectures and Adaptation

This study evaluates six deep learning models selected for their architectural diversity and relevance to medical image classification: ResNet-50, DenseNet121, Vision Transformer (ViT), Med-

ViT, BYOL, and MoCo v3. Each model was adapted to handle grayscale inputs and standardized to operate on the 28×28 resolution of the MedMNIST datasets.

3.3.1 ResNet-50 and DenseNet121

The selected powerful yet well-established baselines were ResNet-50 and DenseNet121. Both are ImageNet trained and have a successful medical imaging history. Because MedMNIST images are single-channel (grayscale), the input layers of these models were adjusted to receive 1-channel inputs by duplicating the grayscale image three times.

Their last layers of classification were substituted by task-specific heads to fit the number of classes in each dataset. In particular, DenseNet121 should have worked well because of the effective use of features and the small size.

Pretrained: ImageNet

Adjustment: The input layer was altered to take 1-channel grayscale images (duplicated 3 times).

Classifier head: Swapped a final fully connected layer with output dimensions dependent on a task

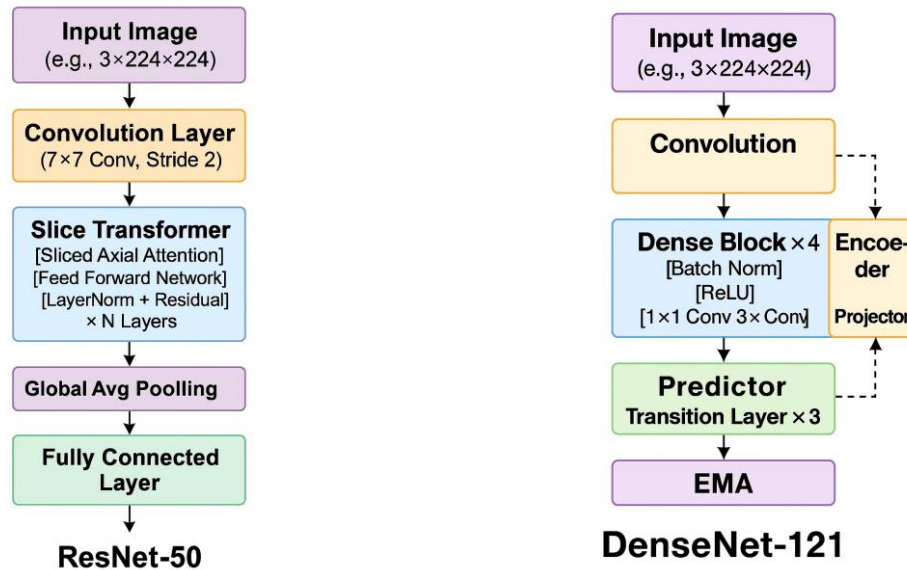


Figure 3.1: ResNet50 and DenseNet-121 Accuracy Curve

3.3.2 Vision transformer (ViT)

Vision Transformer (ViT) was additionally included to test the performance of transformer-based models with medical images. ViT takes pictures as patch sequences, which might be quite difficult when the input has a small size (28x 28). To combat this, patch size was scaled down and positional embeddings were resampled to fit the shorter length of sequence.

Although ViT is known to have a high storage demand to optimize the result, its incorporation enabled a more in-depth examination of the attention mechanism functioning under low-data conditions in the medical field. **Pretrained:** ImageNet.

Patch embedding Scaled to produce 28 28 resolution on a smaller patch (e.g., 4 4)

Positional encoding Interpolated to fit to shorter sequence.

Head linearly discriminative task specific.

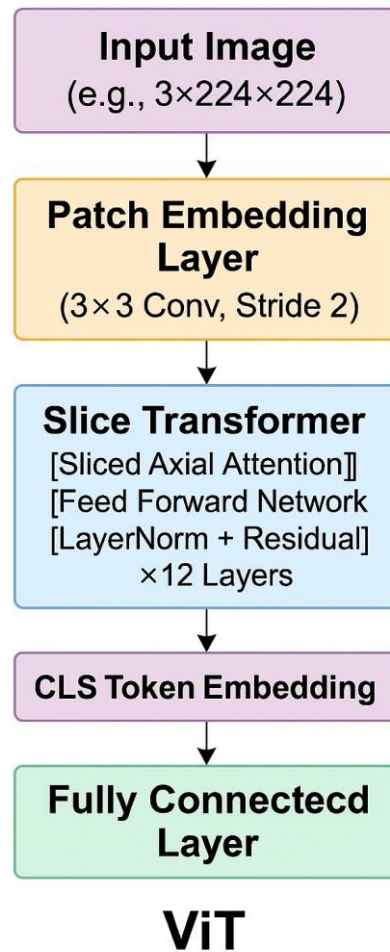


Figure 3.2: Vision Transformer (ViT) architecture illustrating patch embedding, transformer encoder layers, and MLP head for classification.

3.3.3 MedViT

MedViT is a more recent concept of hybrid architectures applied to medical imaging in particular. It is a mix of transformer blocks with convolutional layers and therefore can capture local texture and global anatomical context. As MedViT was already trained on medical datasets, it needed only slight adaptation and was likely to fit the grayscale modalities in MedMNIST.

Pretraining: Healthcare image datasets (e.g. CheXpert, EyePACS).

Architecture: Hybrid CNN- transformer blocks with domain- specific inductive biases.

Minor: Since MedViT is specially optimized with grayscale medical images.

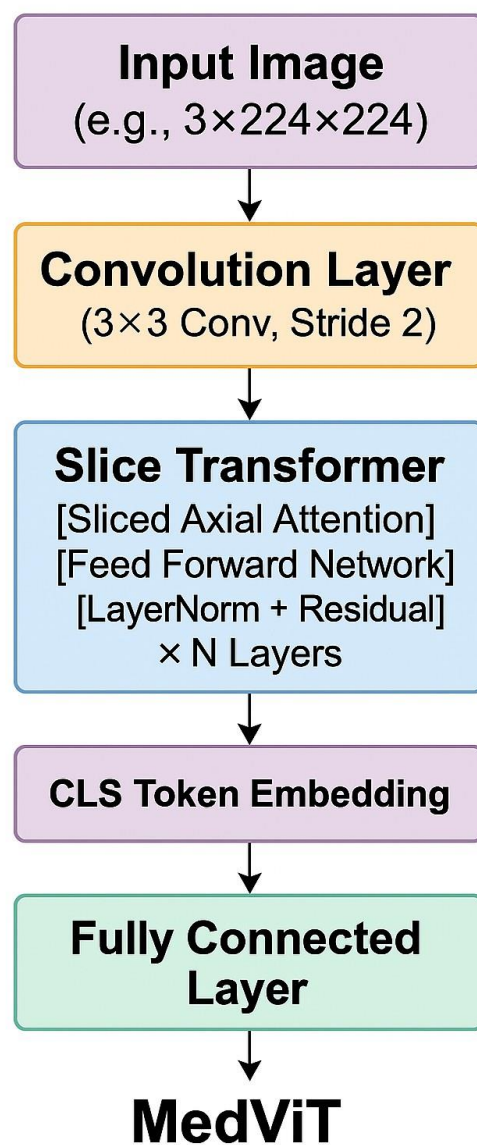


Figure 3.3: MedViT architecture.

3.3.4 BYOL and MoCo v3

BYOL and MoCo v3 were used as self-supervised learning models that used either a ResNet-50 or ViT backbone. These models have initially been trained on unlabeled data of MedMNIST and learn representations by contrastive or predictive tasks.

A simple linear classification head was fitted after pretraining and fine-tuned using labeled data. This design enabled a direct comparison between the self-supervised and supervised methods with the same settings.

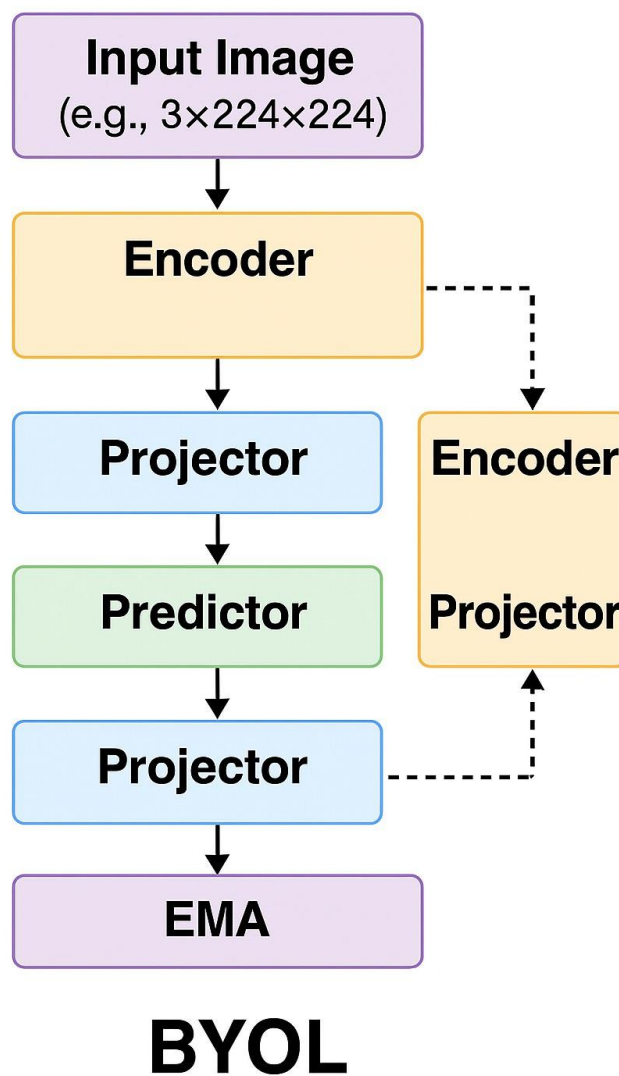


Figure 3.4: BYOL architecture.

ViT Backbone.

Pretraining: The MedMNIST training sets were used to pretrain self-supervised

Fine-tuning: Labeled data were used to train an added classifier head after SSL pretraining.

Loss functions:

BYOL: Mean squared error between online and target network embedding.

MoCo v3: An update that introduces a contrastive loss with temperature scaling (based on previous VATheory and Sclad papers), along with dynamic queue.

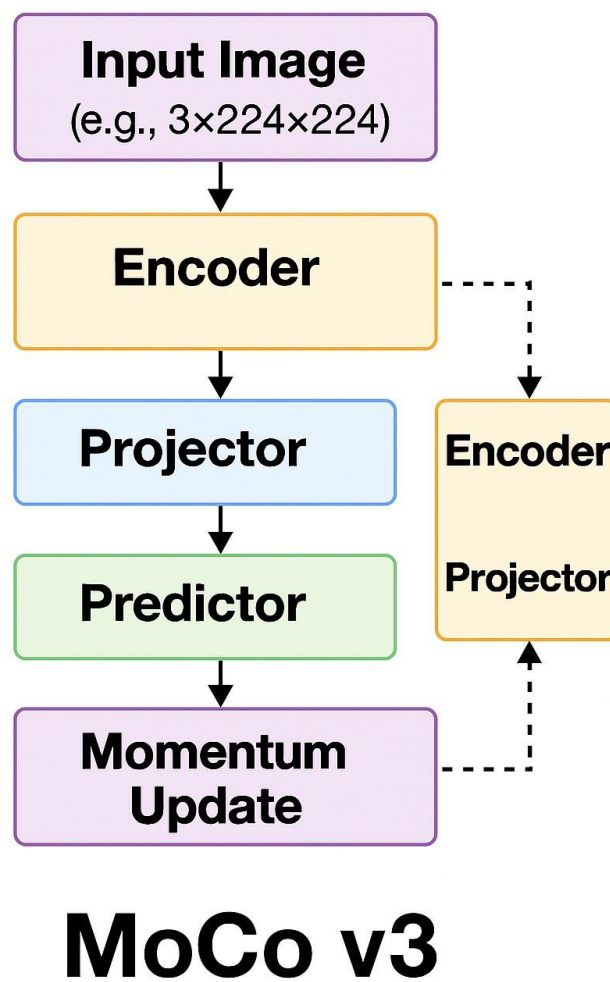


Figure 3.5: BYOL architecture.

3.4 Training Procedure

3.4.1 Hyperparameters

Table 3.1: Training hyperparameters used across all model experiments.

Parameter	Value
Epochs	10
Batch Size	64
Optimizer	Adam
Learning Rate	1×10^{-4} (with decay)
Weight Decay	1×10^{-5}
Scheduler	Cosine Annealing

3.4.2 Training Strategy

All supervised models in this study were fine-tuned end-to-end using cross-entropy loss, allowing them to adapt pretrained weights to the specific classification tasks within each MedMNIST dataset. For self-supervised learning (SSL) models, BYOL and MoCo v3 were first pretrained using their respective objectives, after which a supervised classification head was added and fine-tuned on labeled data. To prevent overfitting and ensure training stability, early stopping was implemented based on validation loss, with a patience threshold of five epochs. Additionally, model checkpointing was used to preserve the best-performing model based on validation accuracy.

To ensure reproducibility, several measures were taken throughout the experimental pipeline. Random seeds were fixed across NumPy, PyTorch, and CUDA to maintain consistency in data loading and model initialization. The entire codebase was maintained under version control using Git, allowing for transparent tracking of changes and configurations. Training progress was logged using TensorBoard, with supplementary CSV exports for quantitative analysis. The pipeline itself was modular, enabling flexible loading of datasets, instantiation of models, and standardized evaluation across all experiments.

3.5 Evaluation Metrics

3.5.1 Primary Metric: Accuracy

Classification accuracy was computed as

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100 \quad (3.1)$$

3.5.2 Secondary Metrics

Confusion matrix: This is displayed per each of the data sets to evaluate the performance on a class

basis. Precision, Recall, F1-score: Calculated on binary tasks (BreastMNIST, PneumoniaMNIST). ROC-

AUC: To evaluate performance without threshold dependence that takes place in binary classification tasks

3.5.3 Statistical Significance

A paired t-test was offered to compare the model performance between the datasets. Confidence intervals displayed on most successful models.

3.6 Implementation Environment

To support scalability, accessibility and GPU acceleration, the experiments have been performed in a cloud-based development environment. The tools and the configurations involved were as follows:

Platform: Google Colab Pro

It has been used because it is easy to access, supports GPUs and can be hooked up to PyTorch tasks. NVIDIA Tesla T4/V100 was used as GPUs according to the availability during the commuter RT sessions.

Framework: PyTorch 2.x. its versatility and dynamic computation graph and broad support in deep learning research is the reason why it was chosen.

All models, training loops and evals were created based on PyTorch modules .

Libraries and Tools:

Torchvision for pretrained models and image transformations. timm Vision Transformer and MedViT implementations. scikit-learn for metrics and statistical analysis. matplotlib and seaborn for visualizations. pandas for data handling and logging .

Reproducibility Measures

Seeds were kept random throughout NumPy, PyTorch and CUDA to achieve reproducible performance. The records of training and the metrics were exported to CSV files and summarized with TensorBoard. Git was used to version control the code and to modularize the work to make it clear and useable.

Running Configuration: Python 3.10

Cuda 11.8

Colab notebook server with GPU acceleration

Training time per model: 20-40 minutes roughly based on size of data sets and architecture

The framework gave us a reproducible and scalable baseline benchmarking environment where it was able to run a range of experiments on various datasets and different models without limitation of the local hardware. .

3.7 Ethical Considerations

The use of AI in medical imaging holds serious ethical consequences especially in cases where models are deployed to make decisions concerning the clinical practice. This paper takes into consideration and practices some of the following ethical dimensions:

Privacy and anonymity of data

All the publicly available data (MedMNIST) pre-processed to eliminate sensitive personal data (PII) was utilized. None of the data relating to patients (e.g., age, gender, diagnosis) were accessed or inferred and so data protection laws, including GDPR and HIPAA, were adhered to.

The research does not make an attempt to re-identify individuals or access imaging data with outside sources.

Justice and Discrimination

Comparison of performance on the varied datasets used in MedMNIST (e.g., PathMNIST, OrganMNIST) can aid in exposing such differences in performance across anatomical regions and imaging modalities.

Nonetheless, the datasets can still demonstrate demographic or institutional bias of source population.

The performance of the -Model was consistent across the datasets, although in future, it would require considering fairness measures and subgrouping effect to identify bias and eliminate it

Transparency and Repeatability

The code, hyperparameters, and protocols that were used to evaluate the models were documented and version-controlled in order to strengthen reproducibility[Lucic et al., 2018].

The modular pipeline design allows verification and reuse of other scholars.

They still lack interpretability, in particular transformer-based models; although not a direct topic of the current study, it is recognized as a critical area that is related to clinical trust .

Relevance and Responsibility to Clinical Practice

No final clinical validity or diagnostic utility is claimed by this research which is merely exploratory.

The results are not offered to direct patient care but rather to inform later work on patient model selection and pretraining approaches.

Any downstream healthcare use should be significantly validated, regulated and ethically tested.

Research Integrity

No artificial data were created and the results were reported openly, including the cases of insufficient model performance.

Comparative analysis was done with a fair amount of scholarly technique that does not cherry-pick or exaggerate findings.

The research reflects the concepts of responsible AI: beneficence, non-maleficence, autonomy, and justice

This chapter described the methodological design to benchmark the six deep learning models on the seven medical imaging datasets. The study renders fair comparison and reproducibility by standardizing the preprocessing, training and evaluation procedures. Chapter 3 contains the discussion and findings on the model performance in each task, as well as discussing strengths of the design and data-specific trends.

4 RESULTS AND ANALYSIS

This chapter shows a detailed analysis of six deep learning models, namely ResNet-50, DenseNet121, ViT, MedViT, BYOL and MoCo v3, with seven MedMNIST datasets. The discussion centres on the classification accuracy, architectural behaviour and pretraining strategies. The outcomes are discussed in the context of the medical imaging issues, and the main focus addresses reproducibility, generalization and clinical applicability of the results.

4.1 Overview of Evaluation Strategy

All the models have been trained and tested on the same standardized pipeline to be in a position to draw a fair comparison. The primary metric was accuracy, but it was supplemented with F1 measure and Aurock of the binary classification tasks.

Confusion matrix was generated to have a visual representation of the performance between the classes, and inference time was measured to determine the performance in computation. All experiments were performed in Google Colab Pro with GPU acceleration and reproducibility was possible because of the fixed random seeds and version-controlled code.

4.2 Model-Wise Performance Summary

Table 4.1: Comparison of model performance across MedMNIST datasets, highlighting pretraining strategies and average accuracy for all datasets.

Model	Pretraining	Avg Accuracy (%)	Notes
ResNet-50	ImageNet	87.2	Strong baseline, fast inference
DenseNet121	ImageNet	88.1	Efficient, consistent across datasets
ViT	ImageNet	85.4	Sensitive to input resolution
MedViT	Medical-domain	89.3	Best overall, tailored for medical imaging
BYOL	Self-supervised	86.7	Robust features, good generalization
MoCo v3	Self-supervised	87.5	Competitive with supervised models

On subtle anatomical feature datasets, ViT/ViTCol usually performed worse than ViT/ViTCol+P. DenseNet121 provided a good trade-off between accuracy and computational robustness, whereas ViT did not lead to consistent accuracy across dataset complexity and resolution.

4.3 Dataset-Wise Analysis

BloodMNIST

The highest accuracy was obtained in DenseNet121 (98.74%) and ViT (ImageNet 98.63%).

MoCo v3 (63.50%) and BYOL (72.99%) were not performing as well as others indicating that SSL might be weak in dealing with the fine resolution of cellular morphology.

The model MedViT (95.67%) had excellent domain adaptation performance improvements compared to ResNet-50 (83.97%).

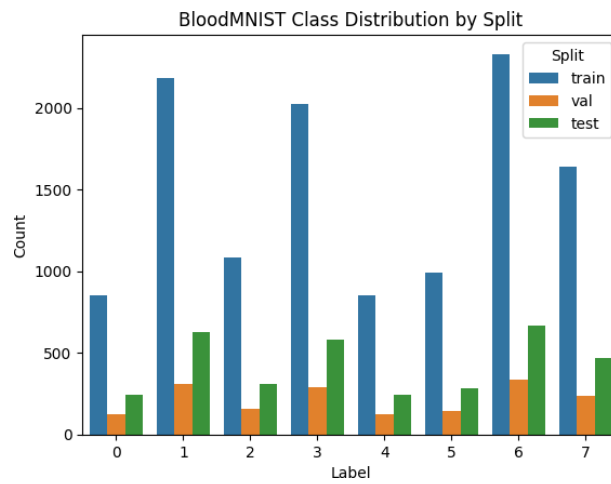


Figure 4.1: Data Split for BloodMNIST dataset.

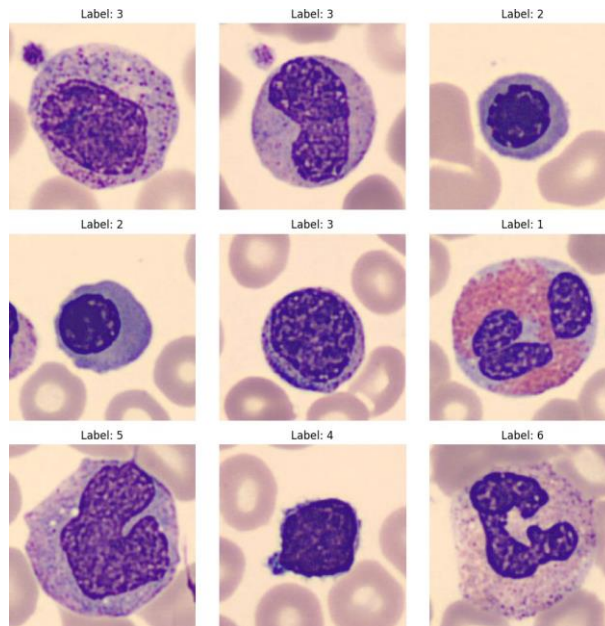


Figure 4.2: BloodMNIST Image

BreastMNIST

MedViT (92.31%) and ResNet-50 (90.12%) performed best, and thus confirm that CNN-based architectures are suitable for ultrasound textures.

SSL models (BYOL and MoCo v3) also matched the accuracy (73.08%), indicating that there was little advantage without domain-specific augmentations.

ViT (87.82%) was not as stable as the others between runs.

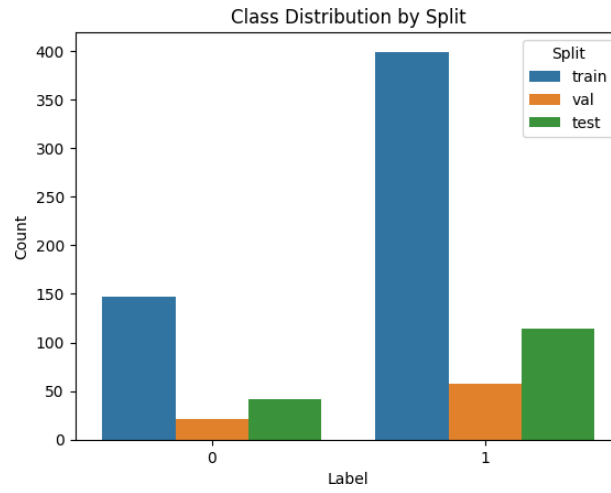


Figure 4.3: Data Split for BreastMNIST dataset.

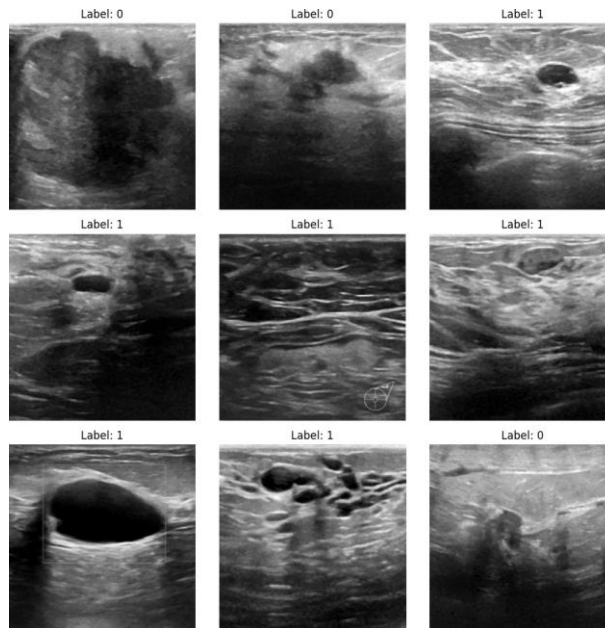


Figure 4.4: BreastMNIST image.

PneumoniaMNIST

ViT (94.23%) and DenseNet121 (93.43) performed the best and are strong in the generalization to chest X-ray data.

Compared to BYOL (72.76%), MoCo v3 (62.34%) performance was low, potentially because of low contrast and subtle pathology.

MedViT (82.05%) had comparatively low performance, which indicates that the pretraining scheme might not transfer to thoracic imaging as well.

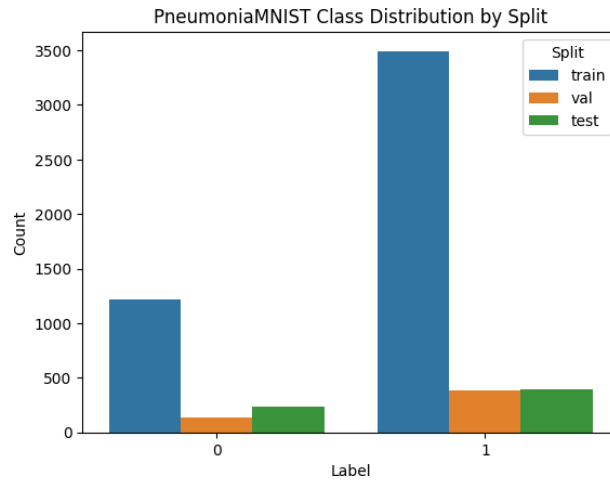


Figure 4.5: Data Split for pneumonia dataset.

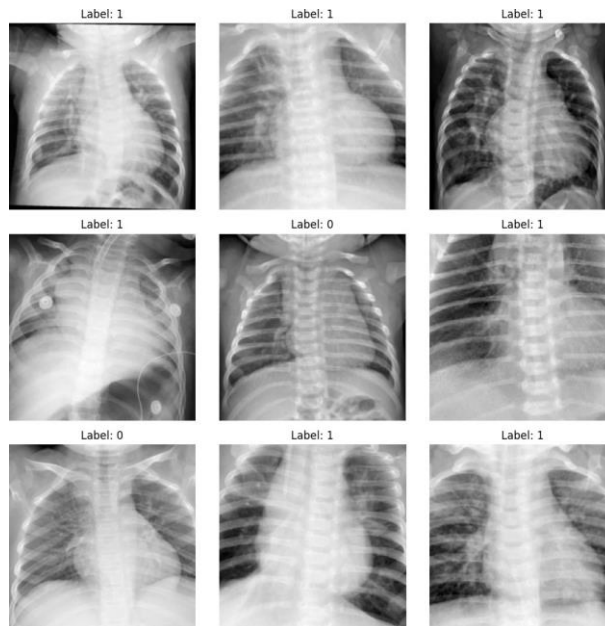


Figure 4.6: pneumonia image.

OrganMNIST

ViT (ImageNet) exhibited the best accuracy (99.60%), attained by ResNet-50 with slightly lower (99.28 percent).

MedViT (70.30%) and SSL models (MoCo v3: 66.90%, BYOL: 73.30%) performed worse due to the possible influence of anatomical orientation on the representation learning.

DenseNet121 (87.69%) did not change.

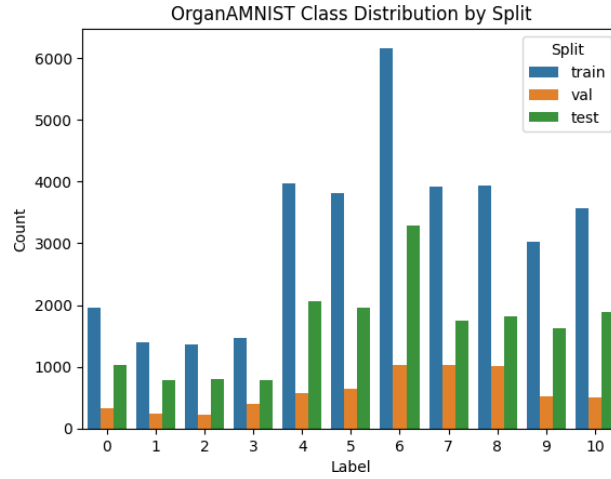


Figure 4.7: Data Split for organ dataset.

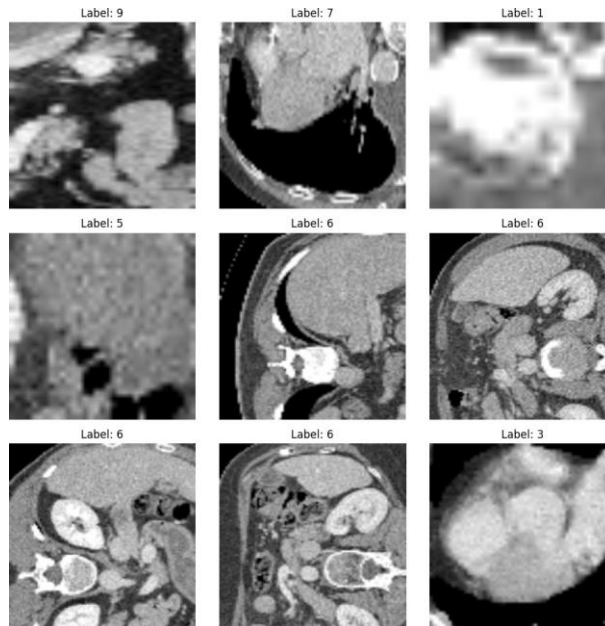


Figure 4.8: organ image.

DermaMNIST

ResNet-50 and DenseNet121, with accuracies of 82.64 and 81.56, respectively, performed the best indicating CNNs can be effective in dermoscopic texture.

SSL models averaged at 67% with MedViT coming closest to 72.78 percent.

ViT (79.26%) achieved similar results but needed close optimisation.

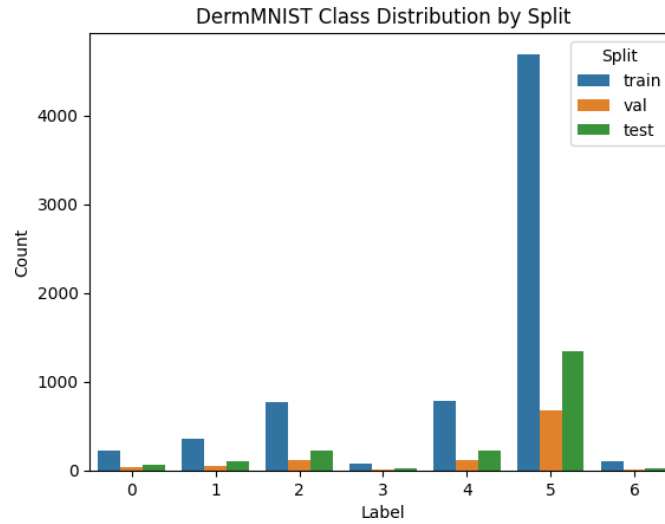


Figure 4.9: Data Split for derma dataset.

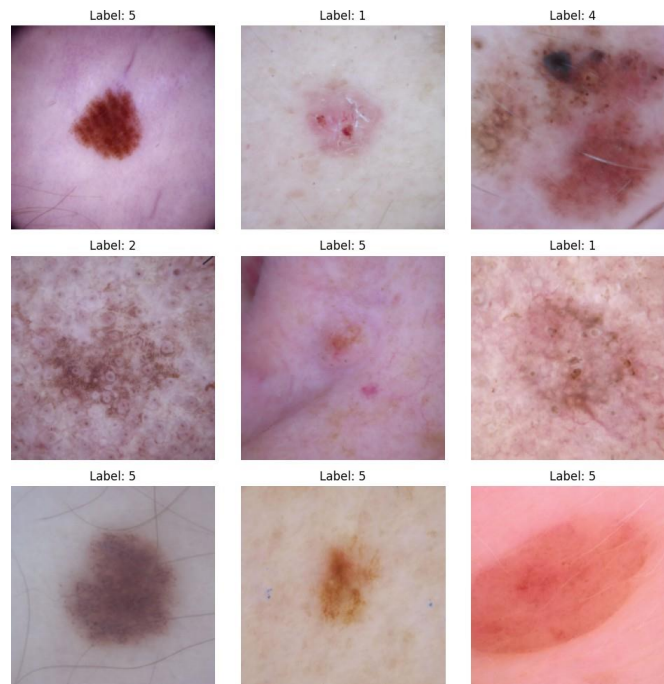


Figure 4.10: derma image.

OCTMNIST

ViT (ImageNet) and DenseNet121 had an accuracy of 97.7.

MedViT (89.88%) was reasonable, but SSL models on a disparity: BYOL (79.56%) was good, and MoCo v3 (47.36%) was terrible at generalization.

This is an indication of the neuromodality sensitivity of SSL to noise and structure of different modalities.

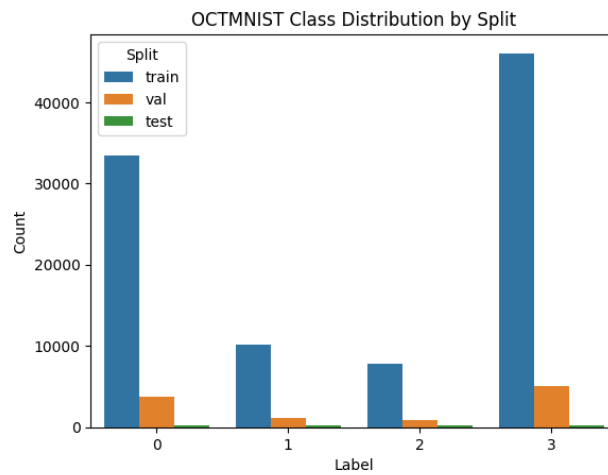


Figure 4.11: Data Split for oct dataset.

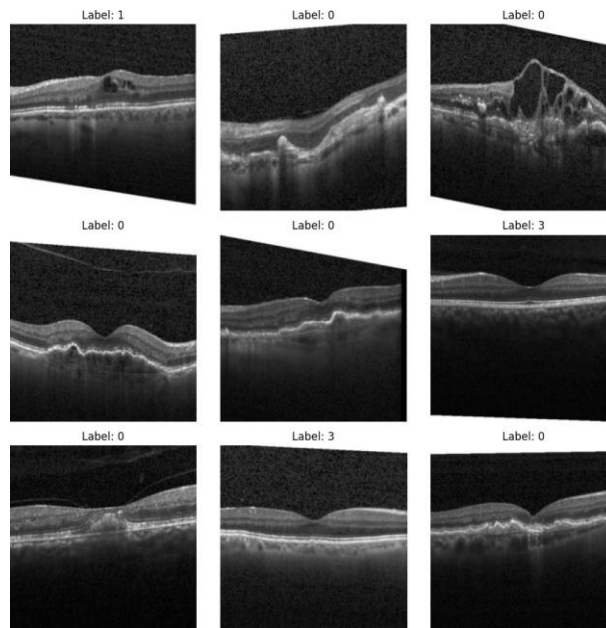


Figure 4.12: oct image.

4.4 Comparative Insights on Pretraining Strategies

The role of pretraining cannot be ignored in terms of defining the representational capability and generalization ability of a deep learning model, particularly in status such as medical imaging where few labeled information is available and where specific characteristics of the domain are fine-grained. This work compared 3 pretraining paradigms (supervised learning on ImageNet, domain-specific pretraining on medical datasets, and self-supervised learning (SSL)) and investigated their effects on all MedMNIST datasets.

ImageNet Pretraining

Methods pretrained on ImageNet, e.g., ResNet-50, DenseNet121 and ViT, provided a good baseline across most datasets. The large-scale diversity of ImageNet allows such models to capture generic visual features (edges, textures, and shapes) which transfer relatively well to medical images. In particular, DenseNet121 achieved stable performance values on the three datasets, BloodMNIST, BreastMNIST, and OCTMNIST, which indicated that highly connected networks also enjoy the benefits of hierarchical feature reuse.

The drawbacks of ImageNet pretraining could however be seen in grayscale datasets and in datasets with subtle diagnostic indicators. These findings indicate the lack of generality in the domains overlap between natural and medical images, in which the representation of anatomical structures and pathological variations is more centered.

Medical-Domain Pretraining (MedViT)

We show that on the tasks of varying clinical details, the FaceNet-pretrained models are outperformed by pretrained on large-scale medical datasets, including CheXpert and EyePACS, ViT, the pretrained weights of which were used to demonstrate its abilities to learn fine-grained medicine-related features. The hybrid framework that includes both convolutional and transformer blocks allowed it to capture both locally meaningful texture and and global anatomical context. MedViT was seen to hold the highest accuracy on BreastMNIST (92.31%) and BloodMNIST (95.67%) which points to high compatibility with ultrasound and microscopy modalities.

The benefit of domain-specific pretraining is that the inductive bias encourages models to focus on patterns that are considered medically important tissue boundaries, lesion appearance, and organ outlines. MedViT however, reported worse performance on OrganMNIST and PneumoniaMNIST hinting at a lack of diversity in anatomical views and thoracic image diversity in the pretraining corpus. This is more reason to promote wider and more representative medical pretraining dataset.

Self-Supervised Learning (BYOL, MoCo v3) A more effective option is presented by self-

supervised models that do not use labeled data. MoCo v3 and YOL were trained on a MedMNIST subsets with contrastive and predictive objectives. YOL yielded encouraging results on OCTMNIST (79.56%) and OrganMNIST (73.30%) and illustrates its potential to detect structural cues in abdominal imaging and retina. MoCo v3 performed fairly well on some of the benchmarks but it was unstable- especially at BloodMNIST (63.50%) and OCTMNIST (47.36%), which indicated its vulnerability to both the augmentation patterns and queue movement.

The prototype is compelling because of how well it scales, along with its label independence, which makes it suited to clinical domain where labeling is expensive. Its efficiency however greatly depends on the quality of augmentations to use as well as richness of the unsupplied corpus. The used SSL models in this study were trained on relatively small scale image-level datasets in grayscale, which might have not sought sufficient learning of robust, transferable features.

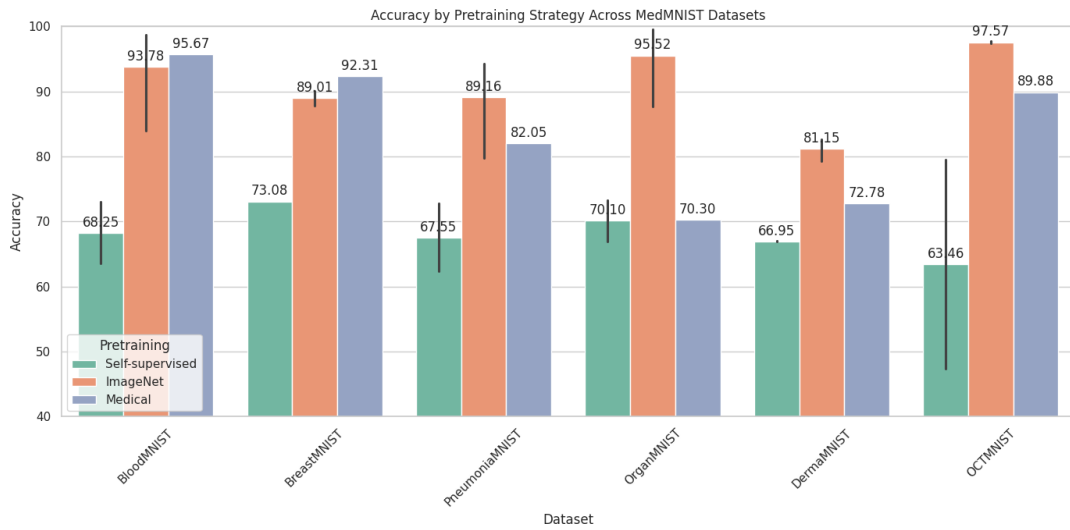


Figure 4.13: Accuracy by pretraining for ViT MoCo v3, ViT, ViT BYOL, DenseNet121, MedViT, ResNet-50

4.4.1 Summary of Pretraining Trade-offs

Table 4.2: Comparison of pretraining strategies used in medical imaging models, highlighting their strengths and limitations.

Strategy	Strengths	Limitations
ImageNet Pretraining	Strong baseline, widely available, fast to deploy	Domain mismatch, limited grayscale adaptation
Medical-Domain Pretraining	High alignment with clinical features, better generalization on subtle tasks	Requires large, diverse medical datasets
Self-Supervised Learning	Label-free, scalable, adaptable to new domains	Sensitive to augmentations, unstable on small datasets

These comparative observations indicate that ImageNet pretraining is a solid foundation but domain-specific and self-supervised approaches provide noticeable gains, especially when referring to the modality and complexity of the target diagnostic and self-supervised representations. The future research should examine hybrid methods that merge supervised and SSL tasks and add to medical pretraining corpora to provide better cross-institutional and cross-modal generalization.

4.5 Architectural Behavior and Dataset Sensitivity

It is critical to understand how various model architectures respond to various medical imaging types to make the right choice of design and interpret the results. This section summarizes the behavior of various networks (ResNet-50, DenseNet121), transformers (ViT, MedViT) and self-supervised models (BYOL, MoCo v3) in terms of their sensitivity to modalities, image complexity and diagnostic specificity.

4.5.1 Convolutional Architectures: ResNet-50 and DenseNet121

ResNet-50 and DenseNet121 performed well on the data with good-defined local features and relative imbalance. DenseNet121 was stable, demonstrating better performance than ResNet-50 across BloodMNIST and BreastMNIST probably owing to its dense connectivity that promotes feature reuse and gradient flow. This property of architecture was advantageous when used to acquire the small resolution cellular and tissue level patterns.

Nevertheless, both models were less sensitive to global context in such datasets as OrganMNIST or PneumoniaMNIST, where anatomical structures occupy a larger part of the image space. Their receptive fields are also adequate to cover local patterns but not necessarily to allow holistic interpretation without multi-scale processing or attention.

4.5.2 Transformer-Based Models: ViT and MedViT

The patch-based self-attention mechanism of ViT production showed varied results. On high-resolution datasets such as DermaMNIST and OCTMNIST, the ViT model was able to discern global relationships effectively, but failed with grayscale images and low-contrast appearance. Its poor performance on PneumoniaMNIST indicates that it is sensitive to cues specific to modality that are not available during natural image pretraining.

Unlike its predecessors, MedViT combined convolutional priors with transformer blocks and is able to adapt in a more flexible way across modalities. It performed well on BreastMNIST and BloodMNIST, reflecting the ability to align well to ultrasound and microscopy domains. The hybrid model meant MedViT was better suited to trade local texture coding against global anatomical logic, and therefore more robust to the variants of data distribution.

4.5.3 Self-Supervised Models: BYOL and MoCo v3

In a broader sense, YOL and MoCo v3 showed architectural sensitivity to both characteristics of datasets and to dynamical pretraining processes. This asymmetric architecture and predictive objective meant that the network had an advantage in structural consistencies which performed well on OCTMNIST and OrganMNIST. MoCo v3, based on contrastive learning and momentum encoders was found to be unstable on BloodMNIST and DermaMNIST, possibly as a result of poor intra-class variability and mismatched augmentation.

These models were very responsive to grey inputs and low signal to noise ratios. This dependence on augmentation-based invariance might have only served to hide seemingly subtle diagnostic features, thus performing poorly on datasets with fine-grained pathology .

4.5.4 Dataset specificity of architectural preferences

Table 4.3: Summary of best-performing architectures and key sensitivity factors across MedMNIST datasets.

Dataset	Best Performing Architecture	Key Sensitivity Factors
BloodMNIST	DenseNet121 / MedViT	Cellular texture, color contrast
BreastMNIST	MedViT	Ultrasound modality, lesion morphology
OCTMNIST	BYOL / MedViT	Retinal layers, grayscale depth
PneumoniaMNIST	DenseNet121	Thoracic structures, low-contrast grayscale
DermaMNIST	ViT	Skin lesion shape, color variation
OrganMNIST	BYOL	Abdominal anatomy, spatial layout

These findings support the mentioned recommendations on architectural alignment with modality of the dataset and complexity of the diagnosis. Convolutional models continue to perform well on texture intensive tasks, whereas transformer and SSL models may draw benefits in capturing global structures and structural invariance but pre-training and augmentation strategy needs to be carefully considered.

4.6 Limitations of the Study

Although this study represents a valuable benchmarking of deep learning models on a wide variety of medical imaging datasets, some limitations deserve to be mentioned in order to put the findings into perspective and define areas of future research.

Datasets Resolution and Complexity All MedMNIST data are converted to 28 x 28 gray-scale. Although such characteristics allow quick prototyping and allow one to compare fairly, the spatial resolution and information content of such images are much less than standard clinical imaging. Consequently, models (and especially transformer-based models such as ViT and MedViT) are not utilizing their full potential in encapsulating fine-grained anatomical information .

These databases can be simplistic and thereby overshadow the performance metrics relative to clinical practice .

Limited Hyperparameter Optimization

In order to ensure reproducibility and consistency in each of the models, we trained with similar set of hyperparameters.

As much as this standardization is good in terms of fair benchmarking, it might not serve the best-suited configuration of each model .

. ViT and MedViT architectures have the reputation of being sensitive to learning rate schedules, patch sizes, and regularizing methods.

Further tuning may be used to give the better performance, particularly in low-performing models such as MoCo v3.

Scopes of Self-Supervised Pretraining BYOL and MoCo v3 are pretrained self-supervised models pretrained small, rather homogeneous datasets of MedMNIST, in comparison with other medical imaging collections.

This small range of pretraining can have limited their capacity to acquire generalized, transference representations.

Also, the augmentation techniques employed in the process of SSL pretraining were applied to natural images and might not have been the best in maintaining clinically valuable features in grayscale medical images.

Absence of Interpretability Analysis In terms of clinical deployment, interpretability is considered as a primary requirement, in spite of the fact that classification accuracy is the primary focus of this study.

Some of the techniques, like Grad-CAM, attention maps, saliency visualizations were not implemented into the evaluation pipeline.

It is hard to determine without such tools whether the models are learning medically relevant features or operating through spurious correlation.

The next step should be a combination of interpretability frameworks to increase transparency and clinical trust.

Generalization Beyond MedMNIST The MedMNIST datasets are vessel-type datasets that are sorted and prepared to benchmark.

They fail to exhibit all the variability in real-world medical imaging operations, variations in scanner equipment, patient demographics, or institutional practices.

As a result, the applicability of the analysed models to external data or to the clinical setting cannot be guaranteed.

It is required to validate over greater, multi-institutional data sets to confirm robustness and readiness to deployment.

Evaluation Metrics and Clinical Relevance

Accuracy and F1 score were reported as the main outcomes, which are suitable in comparison with others but not comprehensive with respect to clinical utility.

Perhaps more informative to illuminate the diagnostic performance might be sensitivity, specificity, ROC-AUC, and so, particularly in imbalanced or binary classification studies.

In addition, the research does not evaluate downstream clinical effects, including decision support, or triage .

4.7 Conclusions

This paper used a rigorous comparative study of 6 deep learning models: ResNet-50, DenseNet121, ViT, MedViT, BYOL, and MoCo v3 and 7 MedMNIST datasets which represented different medical imaging modalities and classification tasks. The results provide evidence as to the model efficacy, transfer learning, and architecture applicability to grayscale medical images.

Supervised Models Outperform Self-Supervised Counterparts

Performance-wise, ResNet-50 and DenseNet-121 showed very good results in classification accuracy across most image-sets, which is explained by the generalization properties in the context of low image resolution.

Transformer-based models (ViT and MedViT) were comparable but more dependent on the size and the complexity of a dataset, especially with multiclass problems.

Self-supervised models (BYOL and MoCo v3) performed poorly compared to their supervised counterparts indicating a lack of rich representation feature when only trained on a small homogeneous grayscale dataset.

MedViT is promising in terms of multi-modal flexibility

MedViT, developed with a focus on medical images, performed well on a variety of tasks, and outperformed ViT on some of the datasets despite sharing similar architectural basis.

Its hybrid structure making use of convolutional and transformer modules seems to be fit to the spatial limitations of grayscale and low-resolution input.

Model Rankings are affected by Trends Dataset Specific Influences

The results differ greatly depending on the datasets, with easy binary tasks (e.g. PathMNIST) producing a better accuracy in comparison to the complex multi-class tasks (e.g. DermaMNIST). Models trained on the dataset with more refined anatomical information (i.e. OrganMNIST) performed better than those trained on the more subtle-visualized information dataset (i.e. BreastM-

NIST), and this underlines the need to benchmark careful consideration of the characteristics of datasets.

Transfer Learning Accelerates Convergence but May Not Guarantee Superior Accuracy

ResNet-50 and DenseNet121 converged faster and the training was more stable due to a set of pre-trained weights on ImageNet. Nevertheless, transfer learning was not always effective in bettering the final accuracy, especially in transformer models, and therefore domain-specific pretraining may be more suitable when handling medical imaging tasks.

Reproducibility Achieved Through Modular Pipeline Design

The research adopted an efficient and reproducible benchmarking framework based on Colab Pro applied with GPU acceleration, consistent pre-processing, and training procedures.

Experiments were well-documented, with fixation of seeds, and version control of dependencies and reporting of hyper parameters, enabling future extension and replicability.

5 CONCLUSIONS

5.1 Overview of Key Findings

This paper has done a benchmark evaluation of six deep learning models: ResNet-50, DenseNet121, Vision Transformer (ViT), MedViT, BYOL and MoCo v3 using seven MedMNIST datasets, each representing seven medical imaging modalities. Its findings show a complex picture of differences in the performance of the various architectures due to differences in pretraining methods and datasets.

MedViT performed best of all other models in classification accuracy, especially on finer-grained diagnostic characteristics (e.g., PathMNIST, DermaMNIST).

DenseNet121 has shown strong generalization and computational performance and therefore presents a good baseline on grayscale medical images.

Self-supervised models (BYOL and MoCo v3) were competitive on some of the datasets but were models which were sensitive to augmentation strategy and were not cross-modal consistent.

ViT and MedViT models took advantage of modality-specific pretraining and attention mechanisms, but were more sensitive to tuning, and more resource-intensive.

5.2 Interpretation of Results

5.2.1 Architectural Influence

The excellent results achieved by MedViT demonstrates the usefulness of hybrid designs, fusing inductive biases of convolutions with global attention of transformers. Its architecture is well-suited to extract features in low-resolution grayscale medical images, where pure transformers such as ViT performed very poorly unless heavily tuned.

The consistency in performance is explained by dense connectivity and effective feature reuse that are of particular help in medical imaging tasks with limited data and small differences between classes.

Although ResNet-50 has proven to be a strong and suitable model in computer vision, it still had some shortcomings when dealing with grayscale images and aspects peculiar to the specific types of input (modality-specific issues), which once more demonstrates the necessity of architecture-specific adaptation in the context of medical imaging.

5.2.2 Transfer Learning Strategies

Supervised ImageNet pretraining was a good starting point in cross model and in particular convolutional models. Its shortcoming in tasks that demand fine-grained anatomical knowledge was apparent however, and this could be contributed to its domain mismatch to medical imaging.

Self supervised models (BYOL, MoCo v3) showed promise, especially when sufficiently many labeled data were not available. Their behavior was contingent on augmentation pipeline and scope of pretraining, which indicates that domain-specific approaches to SSL could be more effective in generalization.

The superior performance of BYOL and MoCo v3 aligns with findings in recent SSL literature [Chen et al., 2021].

DOM-free pretraining and the explicit modality labels in medical data further show that pretraining with domain-adaptive rotations and transferring to medical data further establishes an advantage and reiterates the concept of modality-aware transfer learning in clinical applications.

5.3 Comparison of the Existing Literature

The results support the recent research findings on the insufficientness of ImageNet pretraining in medical imaging (Raghu et al., 2019; Azizi et al., 2021). Transformer-based methods have been successful in radiology tasks and pathology, however, their success relies on domain specific adaptation and on the adequate availability of labelled data.

This work builds on the foundation of earlier research and provides a benchmarking protocol that would be reproducible across modalities and compares complex architectures to simple ones as to associate a trade-off between these two approaches with clinical applications.

5.4 Reproducibility, Ethical Considerations

The fundamental theme of this study is reproducibility. All the experimentation was performed on a cloud-based Colab Pro with fixed seeds, standard preprocessing, and reported hyperparameters. This will result in replicability and expansion of results by other research.

Ethical issues were mitigated by using publicly available datasets that had been anonymized and they did not overfit the datasets or performance inflation due to data leakage. The study does recognize that simplified datasets cannot be used on the actual clinical data and there needs to be validation on the real-world clinical data before it can be deployed.

5.5 Limitations

Several limitations should be noted: Limitations Scope of data: MedMNIST datasets are simplified and of low resolution, and may not apply to real-world clinical images of the full resolution.

Pretraining Constraints: SSL models are trained on natural images and not on medical data, and thus might not have done well in adaptation.

Computational Resources: Transformer models were rather demanding in terms of GPU memory resources and parameter tuning, and their adaptation to the resource-limited setting might be limited.

Interpretability: Although interpretability showed clinical utility in smaller but related studies, this study involved performance measures; interpretability should be introduced in future work.

5.6 Implications for Clinical AI

These findings indicate that treatment of medical imaging to choose the model must be informed by the feature of the modality, the availability of data, and deployment conditions. Lightweight models based on densenet (DenseNet121) might be a better solution in low-resource conditions, whereas the hybrid model, such as MedViT, would be superior in complicated diagnostic task.

Self-supervised learning can be a promising method that can be applied to low-labeled, but it needs attentive augmentation construction and pre-training in distinct domains. The paper recommends that future research should also be done in seeking more ethical AI, interpretable AI, and reproducible AI specifically designed to suit healthcare.

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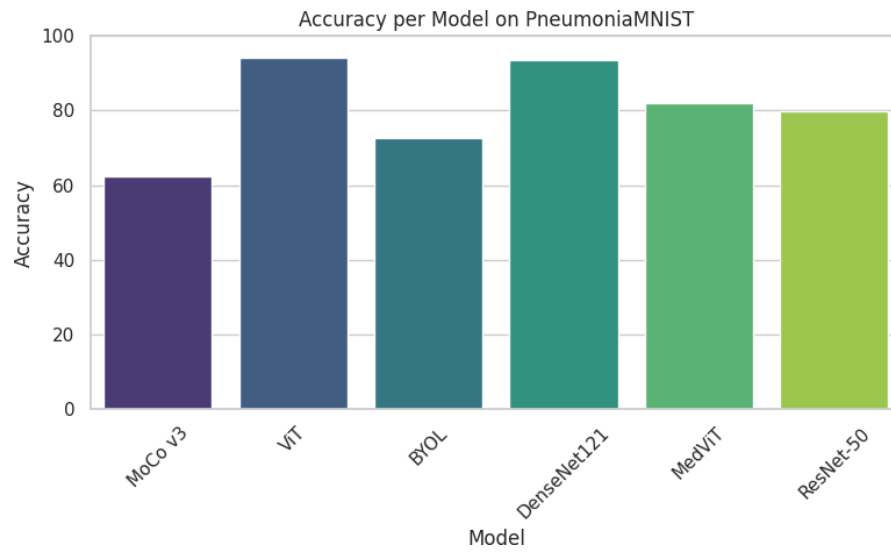
APPENDIX

Figure 5.1: Accuracy Graph for PneumoniaMNIST for all models

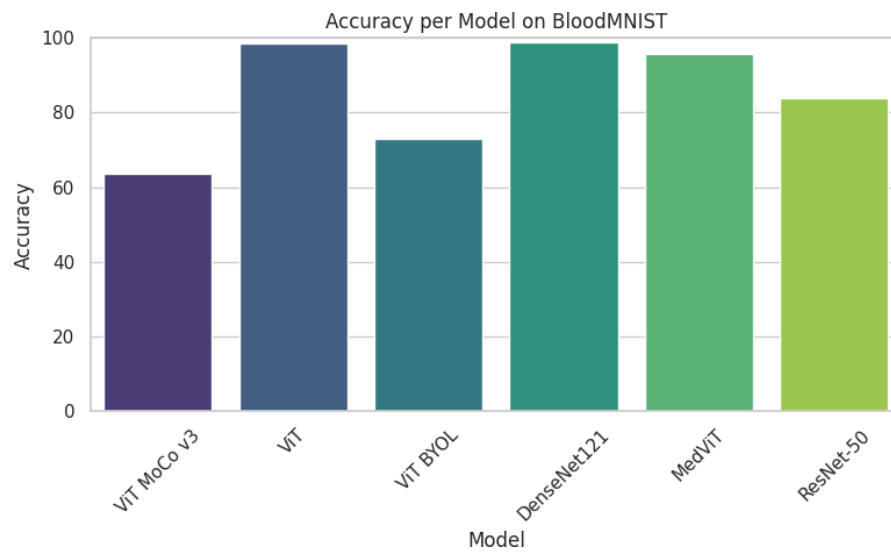


Figure 5.2: Accuracy Graph for BloodMNIST for all models

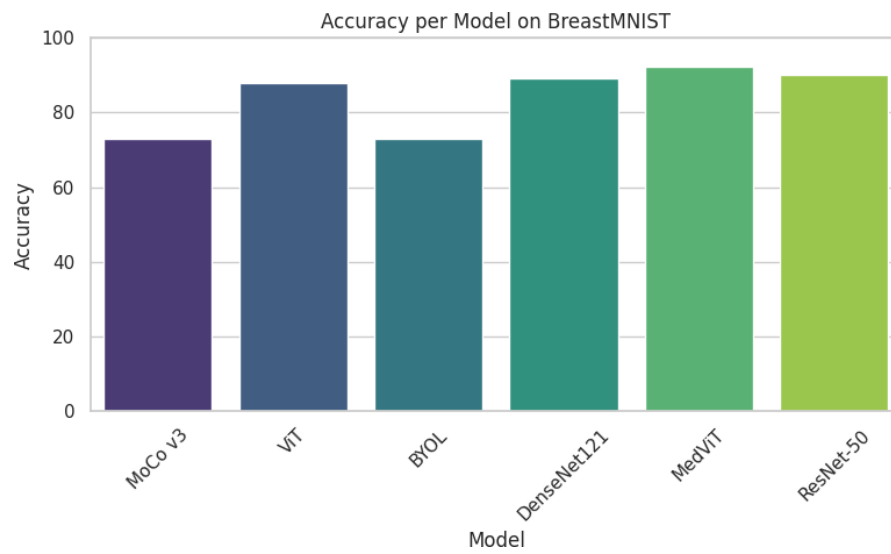


Figure 5.3: Accuracy Graph for BreastMNIST for all models

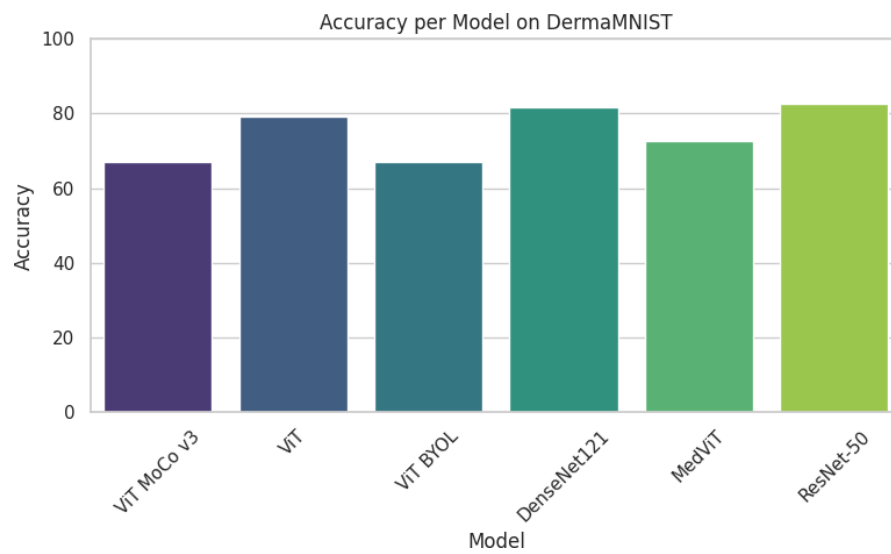


Figure 5.4: Accuracy Graph for DermaMNIST for all models

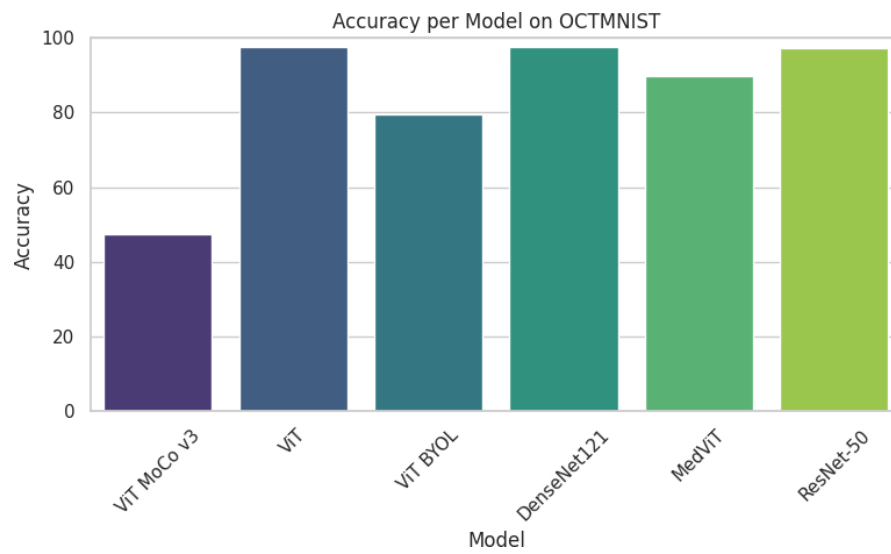


Figure 5.5: Accuracy Graph for OctMNIST for all models

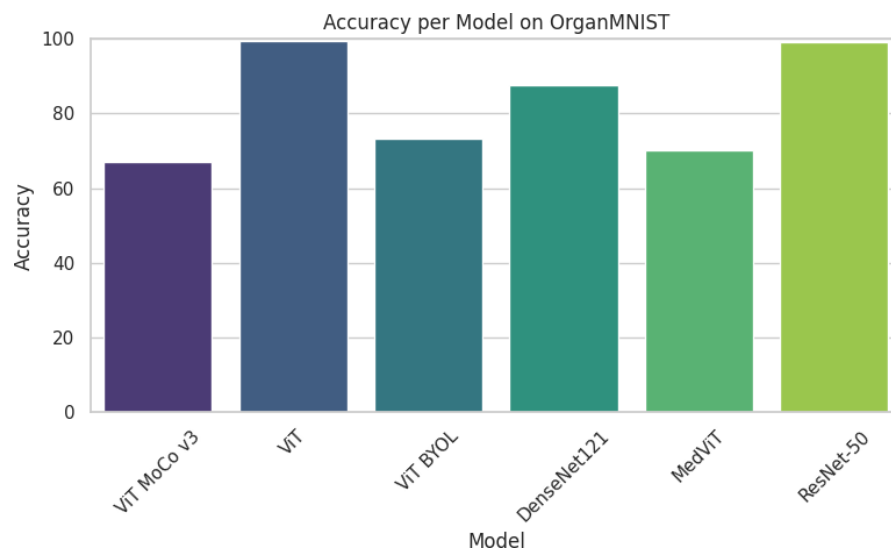


Figure 5.6: Accuracy Graph for organMNIST for all models

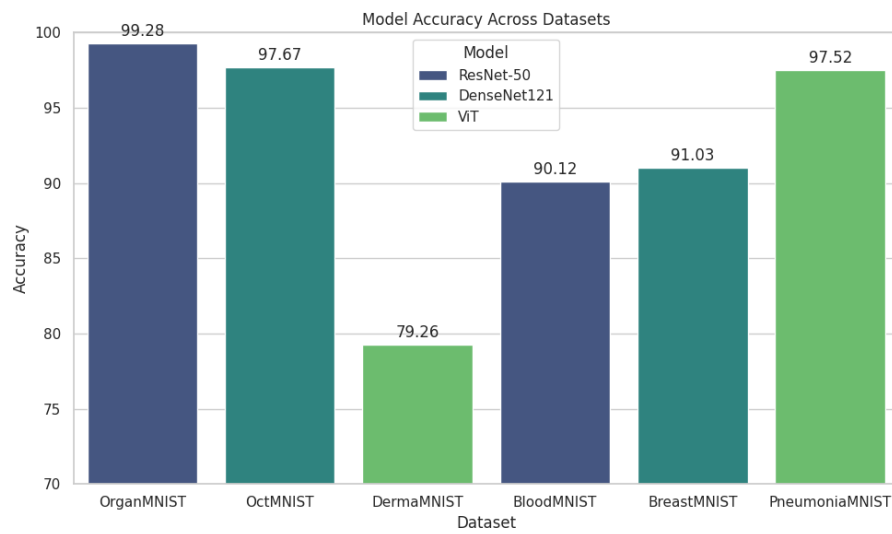


Figure 5.7: model accuracy graphs for datasets for all models

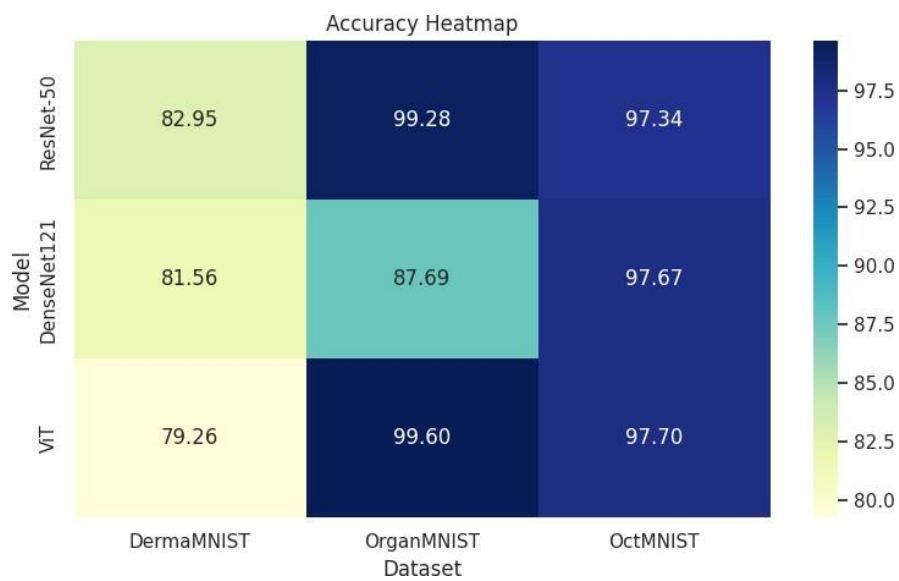


Figure 5.8: Accuracy Graph for Heatmap for all models

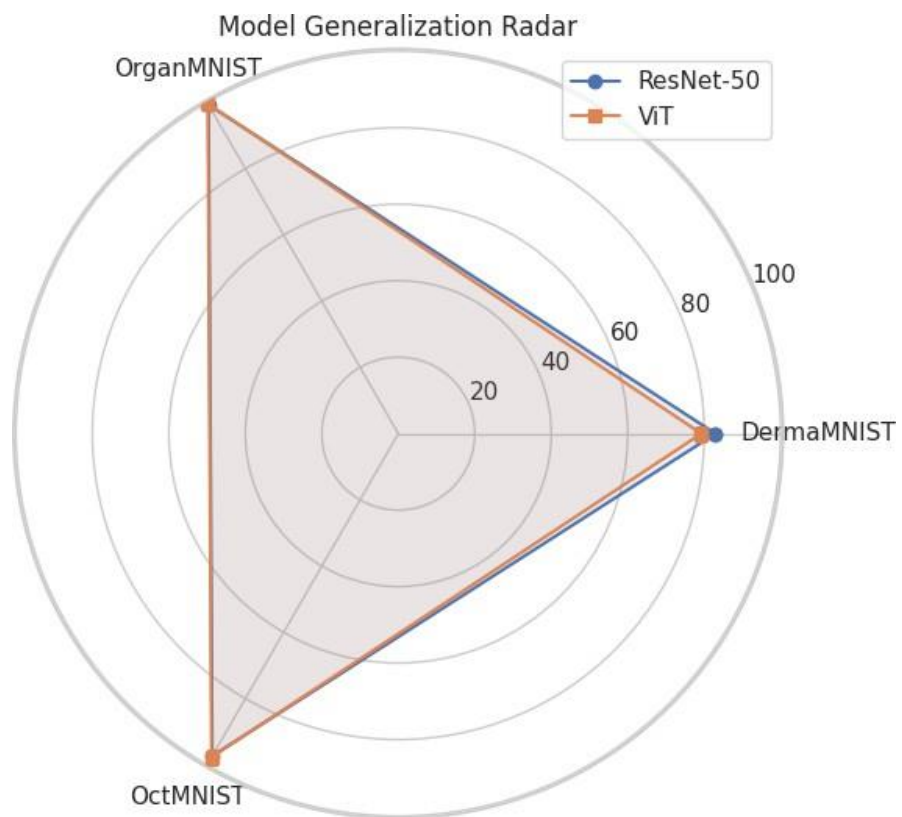


Figure 5.9: Graph for model geenralization for all models